

AI □□ □□ □□ □□□

AI□ □□ □□□□□□□□ □□□ □□□□□



1. **Part 0:** ☐ ☐ ☐ ☐ ☐ ☐ ☐

2. **Part 1:** ☐ ☐ ☐ ☐

3. **Part 2:** ☐ ☐ ☐ ☐ ☐ ☐

4. **Part 3:** ☐ ☐ ☐ ☐ ☐ ☐

5. **Part 4:** ☐ ☐ ☐ ☐ ☐ ☐

Part 0: ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

AI □□ □□□ □□

2023□: "AI□ □□□ □□□□?"

2024□: "AI □□ □□□ □□□□?"

2025□: "AI□ □ □□ □□ VS □ □□ □□"

□□ □□□

AI □□ □□□ □□ □□□ □□ □□

□□□ □ □□ □□□ □□□ □□□ □□□□ □□□□

□ □□□□ □□

1. AI □□ □□□ □ □□□□□ □□
2. AI□ □□ □□ □□□ □□□□□ □□
3. □□□□ □□□□ □□ □□
4. □□ □□□ □□□ □□

Part 1: □□ □□

"□□□ □ □□□□□ □□□ □ □ □ □□"

1.1 토큰 (Token)

LLM은 입력을 "토큰" 단위로 분할하여 처리

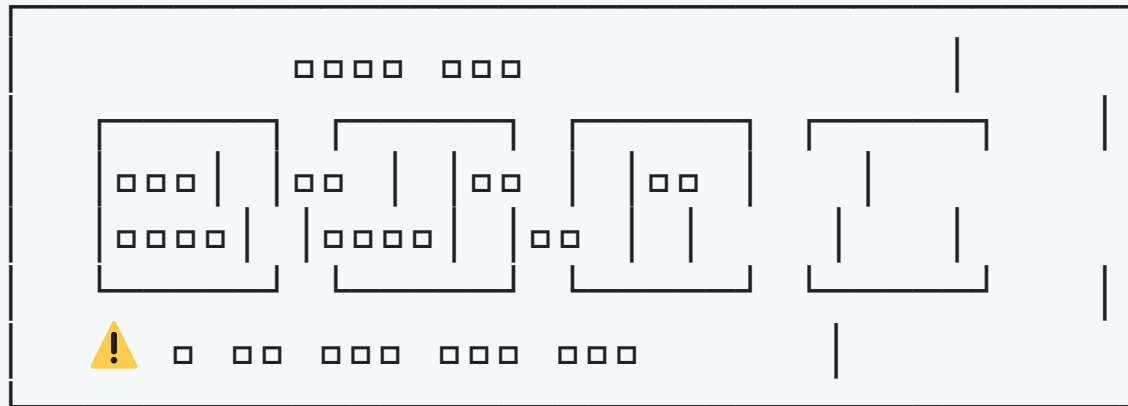
"Hello, world!" → ["Hello", ",", " ", "world", "!"] (4 토큰)
"□□□□□" → ["□", "□", "□", "□", "□"] (5 토큰)

토큰	토큰
토큰	토큰 = 토큰. 토큰 토큰 ~3 토큰 토큰 토큰 토큰
토큰	토큰 토큰 토큰 토큰 토큰 토큰
토큰	토큰 토큰 토큰 토큰 토큰 토큰

1.2 □□□□ □□□

LLM□ □ □□ "□ □ □□" □□□□ □□ □□

□□: □□ □ □□ □□



LLM 比較表

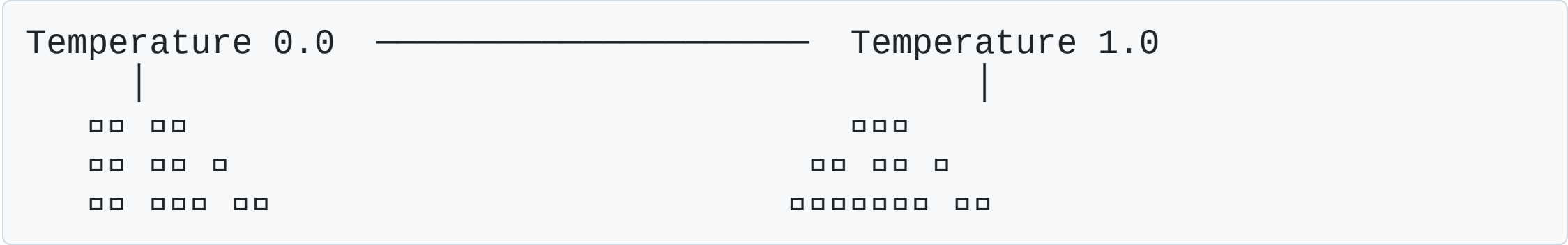
モデル	コンテキスト長	特徴
GPT-3.5	16K tokens	A4 紙 20枚
GPT-4	128K tokens	約 100 頁
Claude 3.5	200K tokens	約 150 頁
Gemini 2.5	1M tokens	約 750 頁

LLM の

- LLM は AI の 未来 を 予測 できる ？ → 未来 予測
- LLM : 未来 予測 能力 向上 , 社会 への 影響

1.3 Temperature (□□□ □□□)

AI □□□ "□□□□"□ □□□□ □□



□□	□□ Temperature
□□ □□	0.0 ~ 0.3
□□ □□	0.0
□□□□ □□	0.7 ~ 1.0
□□ □□	0.3 ~ 0.5

1.4 □□ □□: RAG

Retrieval-Augmented Generation

□□: □□□ □□

□□ LLM: "□□ □□ □□□□ □□□" (□□ □□)
RAG □□ LLM: "□□ □□ □□□□ □□□" (□□□ □□)

□□ □□:

□□ → □□ □□ □□ → □□ □□ + □□ → LLM → □□

1.4 □□ □□: Function Calling

Tool Use

□□: □□□□ □□□

```
□□□: "□□ □□ □□□"
```

```
LLM (□□□): "□□ □□ □□ □□. □□ API □□□□□□"
```

↓

```
Tool (□□): get_weather("□□") → {"temp": 15, "condition": "□□"}
```

↓

```
LLM: "□□□ □□ 15□□□ □□□□"
```

1.4 □□ □□: Agent

□□: □□□□ □□

□□ LLM: □□ → □□ (1□□)

Agent: □□ → □□ → □□ → □□ → □□ □□ → ... → □□

Agent□ □□ □□ (ReAct)

1. Think: "□□□ □□ □□?"
2. Act: □□ □□ (□□ □□, □□, □□ □□)
3. Observe: □□ □□
4. Repeat: □□ □□□□ □□

1.4 □□ □□: MCP

Model Context Protocol

□□: USB-C □□

Before MCP: □ □□□ □□ □□ □□ (□□)

After MCP: □□□□ □□ □□ (□□□)

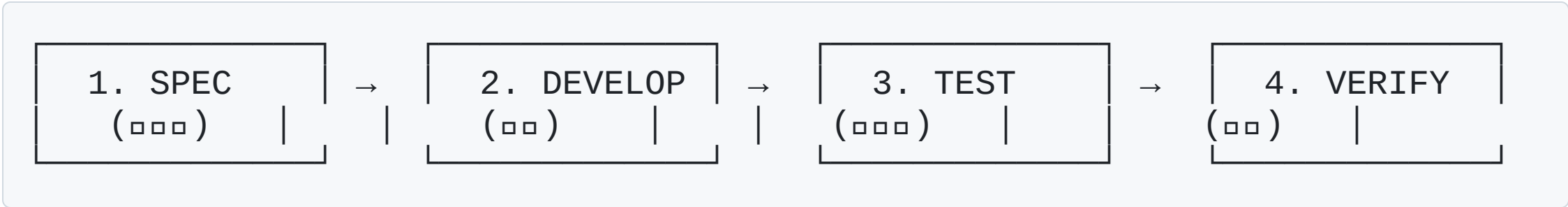
AI□ □□ □□□ □□□□ □□ □□□□ (Anthropic □□)

- □ □ □□ MCP □□□ □□ MCP □□ AI□□ □□ □□
- □□ □□□□ AI□ □□ □□ □□

Part 2: □□ □□□□□

"AI□ □□ □ □□□ 4□□"

□ □ □ □ □ □ □



2.1 SPEC ()

"AI "

AI . .

Human	AI

□ □ □ □ □ □

- ## 1. □ □ (Goal)
- ## 2. □ □ □ (Non-Goals) ← □ □ □ □ !
- ## 3. □ □ □ □ □
- ## 4. □ □ □ □ □ □
- ## 5. □ □ □ □ □
- ## 6. □ □ □ □ □

□□ □□ **vs** □□ □□

✗ □□ □□

```
□□□□ □□□□ □□□ □ □□□ □
```

✓ □□ □□

```
## □□□ □□□□  
### □□ □□: PDF (A4, □□), CSV (UTF-8)  
### □□ □□: 10,000□ □□ 5□ □□ □□  
### □□ □□□: □□□ □□ → □□□ □□
```

2.2 DEVELOP (□□)

□□ □□

"AI□ □□□ □□□"

□□□ □□□ □□□□□□□, □□□□ □□□ □□□□ □□□□ □□.

□□ □□

Human	AI
□□□□ □□	□□ □□
□□ □□	□□□□□□□□ □□
□□ □□□	□□ □□

□□□□ □□□□: CLAUDE.md

```
# □□□□ □□  
Next.js 14 App Router + TypeScript + Tailwind CSS  
  
# □□ □□□  
- □□□ □□□□□ □□  
- Props□ interface□ □□  
  
# □□ □□  
- any □□ □□ □□  
- console.log □□□□ □□□ □□
```

□□ □□

1. □□□ □□ □□□ □□
└ "□□ □□ □□□□" (✗)
└ "□□□ □□ validation □□□ □□□□" (✓)

2. □□ □□ □□ □□□□
└ "□ □□□□□ □□ □ □□□□□ □□□□"

3. □□ □□ □□
└ "□□ API □□□□□□ □□□□ □"

4. □□□ □□ □□
└ □□ □□□ □□ □□ □□

2.3 TEST (□□□)

□□ □□

"AI□ □□ □□□□ □□ □ □□"
□□□□ □□□ □□□ □□□ □□□□□ □□□ □□□□ □□.

□□ □□

Human	AI
□□□ □□ □□	□□□ □□ □□
assertion □□	□□ □□□ □□
□□□ □□ □□	□□□□□□□ □□

TDD with AI

1. Human: `□□□□□ □ □□ □□`
`"□□□ □ □□□ □□□□□ □□ □□□ □□"`
2. AI: `□□□ □□ □□`
`test("shows error on wrong password", ...)`
3. Human: `□□□ □□ □□`
`"□ □□□□ □□ □□ □□□□ □□□□?"`
4. AI: `□□□ □□□□ □□ □□`
5. CI: `□□□ □□ + mutation testing`

2.4 VERIFY (□□)

□□ □□

"AI □□□ **1**□ □□, □□ □□□ □□"

AI□ □□□ □□□ □□□ □□, □□□ □□□□ □□□ □□□ □□□□.

□ □ □ □ □

Layer 1: AI □ □ □ □

- □ □ □ □, □ □ □ □, □ □ □ □



Layer 2: Human □ □

- □ □ □ □ □ □, □ □ □ □



Layer 3: CI □ □ □

- □ □ □ □ □ □, □ □ □ □, □ □

Part 3: □□□□ □□

"□□ □□□□ = □□ □□"

3.1 字符串

字符串	字符串	字符串
字符串	"字符串"	"Line 42 null 字符串"
字符串	"字符串"	"public API 字符串"
字符串	"字符串"	"1: 字符串, 2: API, 3: UI"
字符串	(字符串)	"字符串"
字符串	(字符串)	"JSON字符串"

3.2 □□ □□ □□□

```
## □□  
[□□ □□] VS [□□ □□]
```

```
## □□ □□□  
[□□□ □□ □□□ / □□ □□□□]
```

```
## □□ □□  
[□□ □□□□]
```

```
## □□ □□□ □  
- □□ 1: [□□]
```

```
## □□  
1. □□ □□ □□□ □□□□  
2. □□□□ □□□□ □□□□
```

3.2 □ □□ □□ □□□

```
## □□  
[□□ □□]
```

```
## □□ □□  
- [□□ 1]  
- [□□ 2]
```

```
## □□□ □□ □□  
[□□□ □□ □□ □□]
```

```
## □□  
1. □□ □□ □□□ □□□□  
2. □□ □□□□ □□ □□□□  
3. □ □□ □□ □□, □□□□ □□□□
```

3.3 □□ □□

Chain of Thought (□□□ □□)

```
□ □□ □□□ □□□:
1. □ □□ □□ □□□□ □
2. □□ □□□ □
3. □ □□ □□□ □
4. □ □□ □ □□ □ □□ □

□ □□ □□□□ □□.
```

3.3 □□ □□

Few-shot (□□ □□)

```
□□ □□□□ □□ □□□□ □□□□:
```

```
□□ 1:
```

```
□□: □□□ □□ □□
```

```
□□: "□□□ □□□ □□□ □□□□□□."
```

```
□□ 2:
```

```
□□: □□□□ □□ □□
```

```
□□: "□□□□□ □□ 8□ □□□□□ □□□."
```

```
□□ □□□□:
```

```
□□: □□□□ □□ □□
```


3.3 □□ □□

□□ □□ (Role Prompting)

```
□□ 10□□ □□ □□ □□□.
□□ □□ □□ □□□,
□□ □□ □ □□□ □□.
```

```
□□ □□ □□□:
[□□]
```

Part 4: □□ □□□ □□

4.1 AI □□

□□	□□	□□□
AI □□ □□□ □□	□□, □□ □□□	□□ □□ □□ □□
"AI□ □□□ □□□□"	□□ □□ □□	□□ □□□ □□□

□□□ □

AI□ "□□□□□ □□ □ □□" (Confidently Wrong)

4.2 □□□□ □□

□□	□□	□□□
□□□ □□□	□□□ □□	□□, □□□□ □□ □□
□□ □□□ □□	□□□ □□	□□□ □□ □□

□□ □□□□ □□□□□

- [] □□□ □□□ □□?
- [] □□□ □□□□?
- [] □□□ □□ □□□□?
- [] □□ □□ □□□ □□?

4.3 □□ □□ □□

□□	□□	□□□
"□□ □ □□□□"	□□ □□	□□ □□□ □□
□ □□ □□ □□ □□	□□□ □□□	□□□ □□□□□

□□ □□ □□

✗ "□□□ □□ □ □□□"

✓ "□□□ □ □□ validation □ □□□"

4.4 □□ □□ □□

□□	□□	□□□
□□□ □□	□□ □□	"□ □□□ □□?" □□□□
□□ □□ □□	□□□ □□	□□ □□ □□

□□□□ **AI** □□□

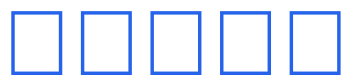
1. AI□ □□ □□
2. "□ □□□ □ □□□ □□□□"
3. "□ □ □□□ □□□□?"
4. □□□ □ □□

4.5 □□ □□

□□	□□	□□□
API □ □□□□□ □□	□ □□	□□ □□ □□
□□ □□ □□	□□ □□	□□ □□ □□ □□

□□ □□□□□ □□□□ □ □

- API □, □□□□
- □□ □□□□
- □□ □□ □□



□□ □□ □

- [] □□ □□ □□ □□
- [] □□□ □□□
- [] □□ □□□ □□□
- [] □□□□ □□□□ □□ □□ (CLAUDE.md)

□□ □

- [] □□ □□□ □□
- [] □□ □□ □□
- [] □□□ □□ □□ □□
- [] □□ □□□ □□□ □□

□□□ □

- [] □□□ □□ □□
- [] assertion□ □□□ □□□□□ □□
- [] □□ □□□ □□
- [] CI □□

□□ □

- [] □□ □□ □□ □□
- [] □□ □□ □□
- [] □□ □□
- [] □□ □□□□



AI ☐☐ ☐☐☐ ☐☐☐☐ ☐☐ ☐☐☐☐ ☐☐☐ ☐☐☐.

□ □ □ □ □ □ □ :

1. ☐☐☐ ☐☐☐☐ (Part 1)
2. ☐☐☐☐ ☐☐☐☐☐☐ ☐☐☐ (Part 2)
3. ☐☐☐☐ ☐☐☐☐☐ ☐☐☐☐ (Part 3)
4. ☐☐ ☐☐☐ ☐☐☐ ☐☐ (Part 4)



"AI □ □ □ □ □ □ ."

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ . "

□ □ □ □ □

□ □ □ □

- □ □ □ □ □ □ □ □ □ □ □ □: README.md
- Marp: <https://marp.app>

□ □ □

```
# PDF □ □  
npx @marp-team/marp-cli presentation.md --pdf  
  
# HTML □ □  
npx @marp-team/marp-cli presentation.md --html  
  
# PPTX □ □  
npx @marp-team/marp-cli presentation.md --pptx
```